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Heterogeneous Graph Neural Network with Adaptive Relation Reconstruction

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ABSTRACT

Topological structures of real-world graphs often exhibit heterogeneity involving diverse nodes and relation types. In recent years, heterogeneous graph learning methods utilizing meta-paths to capture composite relations and guide neighbor selection have garnered considerable attention. However, meta-path based approaches may establish connections between nodes of different categories while overlooking relations between nodes of the same category, decreasing the quality of node embeddings. In light of this, this paper proposes a Heterogeneous Graph Neural Network with Adaptive Relation Reconstruction (HGNN-AR²) that adaptively adjusts the relations to alleviate connection deficiencies and heteromorphic issues. HGNN-AR² is grounded on distinct connections derived from multiple meta-paths. By examining the homomorphic correlations of latent features from each meta-path, we reshape the cross-node connections to explore the pertinent latent relations. Through the relation reconstruction, we unveil unique connections reflected by each meta-path and incorporate them into graph convolutional networks for more comprehensive representations. The proposed model is evaluated on various benchmark heterogeneous graph datasets, demonstrating superior performance compared to state-of-the-art competitors.

1. Introduction

Graphs are commonly encountered in real-world applications, such as citation networks, purchasing behaviors, etc, where nodes and edges represent entities and their relationships. Unlike the inherent geometric and sequential structures observed in images and text, graphs exhibit non-Euclidean data properties, which enable them to capture more flexible and intricate relationships. This distinctive nature of graphs presents certain challenges for the direct application of traditional machine learning techniques. In recent years, machine learning-based approaches have been increasingly extended to better accommodate graph-structured data. Graph Neural Networks (GNNs) leverage the principle of local connections and weight sharing to aggregate and propagate messages among nodes, facilitating an inference of node and edge representations (Kipf & Welling, 2017; Wu, Lin, Lin, Chen, Bai, & Wang, 2023). GNNs are trainable and deployable across diverse types of graphs, providing wide-ranging applications in social networks (Hu, Cao, Hu, & Zhang, 2024; Shen, Jiang, et al., 2023), recommendation systems (Cai, Qian, Fang, Hu, Ding, & Xu, 2023; Han, Hu, Li, Qi, Tang, & Guo, 2024), and knowledge graphs (Che & Tao, 2024; Jiang, Zhou, Zhao, & Wen, 2023).

Owing to the diversity of entity types, numerous real-world graphs gradually manifest heterogeneous characteristics, frequently encompassing multiple entity types (i.e., nodes) and different interactions (i.e., edges). Corresponding methods to accommodate different types of nodes and edges are urgently considered to handle information-rich heterogeneous graphs efficiently. Meta-path (Fu, Zhang, Meng, & King, 2020; Ji, Wang, Shi, Wang, & Yu, 2023; Sun, Han, Yan, Yu, & Wu, 2011), which represents composite connections between two entities, is widely employed for semantic representation in heterogeneous graphs. The meta-path can be conceived as a vehicle to articulate high-order proximity between two nodes. Methods based on meta-paths exhibit the ability to capture intricate interconnections among diverse entities on heterogeneous graphs, extracting abundant semantic relations. By selecting different types of meta-paths, we can capture multifarious relations between nodes, aiding the model in comprehensively understanding the structures of heterogeneous graphs. Most heterogeneous graph learning methods explore node embeddings based on manually designed meta-paths with differing lengths and aggregating node information from their discovered neighbors (Chen, Wu, et al., 2024; Liu, Song, Gao, & Shang, 2022; Liu, Song, Wang, & Shang, 2023).

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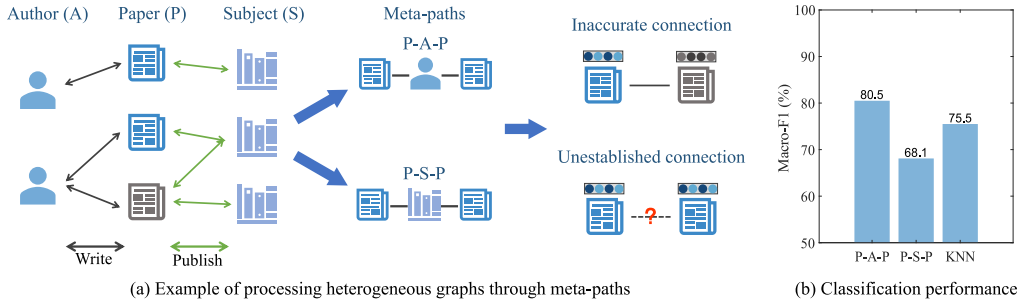


Fig. 1. An illustrative example of a heterogeneous graph: (a) A simple citation network consisting of three types of nodes, employing different meta-paths (e.g., P-A-P and P-S-P) to construct links; (b) Employing meta-paths leads to inaccurate connections and missing connections. Node classification is performed on various meta-paths and on matrices constructed using K-Nearest Neighbors (KNN) respectively.

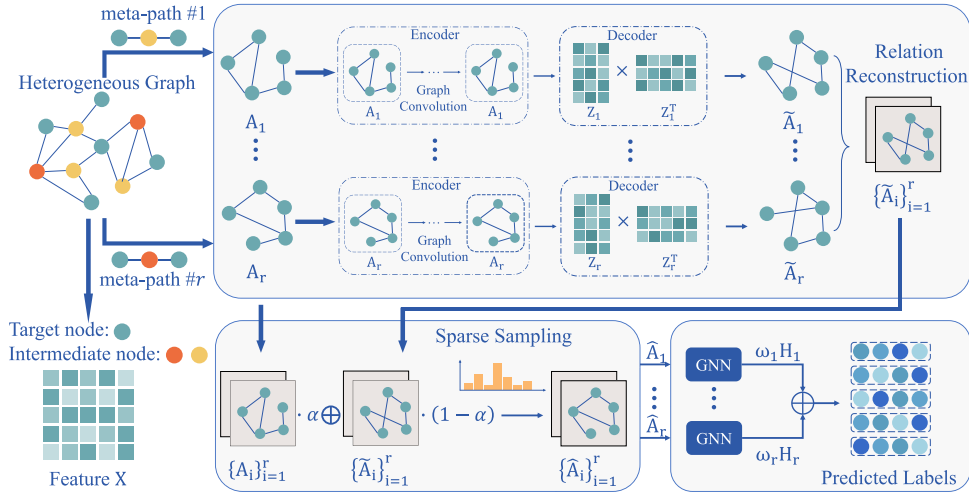


Fig. 2. The overview of the proposed framework. The model explores the connections between nodes in the graph through diverse meta-paths. The model first constructs adjacency relations from different meta-paths to explore the potential feature homogeneity among nodes and establishes accurate connections reflective of the meta-paths. Subsequently, the model employs graph convolutions for each reconstructed unique meta-path adjacency relation to generate representations and perform fusion.

As depicted in Fig. 1, a citation network illustrates relationships among papers through meta-paths such as PAP (Paper–Author–Paper) and PSP (Paper–Subject–Paper). The PAP meta-path suggests that papers authored by the same individual are more likely to belong to the same field, while the PSP meta-path differentiates the field of a paper based on its subject. Although meta-paths are designed to facilitate connections between papers within the same field, they may inadvertently establish links between different types of papers due to noise and the complexity of heterogeneous graph structures, while overlooking relationships between papers with similar characteristics. Likewise, this observation holds across different meta-paths. Furthermore, constructing node relationships based on features, such as through K-Nearest Neighbors, tends to result in insufficient or noisy connections, which can similarly result in performance degradation. In scenarios where inaccurate connections exist in the adjacency matrix, graph neural networks encounter a difficulty where initially homogeneous nodes are influenced by inter-class edges during message aggregation and propagation. The disparities between features progressively intensify. This phenomenon indicates that inaccurate connections disturb the uniformity of message propagation. Under such circumstances, the amplification of differences during the message propagation may negatively impact the model performance, leading to biases in the node classification.

To address the aforementioned limitations, we propose a novel model named Heterogeneous Graph Neural Network with Adaptive Relation Reconstruction (HGNN-AR²), which adaptively adjusts the connection by meta-path construction to mitigate adjacency relation deficiency and heterogeneity. As shown in Fig. 2, HGNN-AR² leverages

distinctive connections derived from multiple meta-paths to capture homogeneous patterns, thereby refining precise relationships, uncovering latent associations within complex graph structures, and constructing topological relationships with rich semantics. In addition, we synthesize the enhanced topology with the original structure and introduce a sparsification operation to preserve the homomorphic relations and reduce the interference of irrelevant information. We reveal the unique connections reflected by each meta-path and introduce them into graph convolutional networks to generate more comprehensive representations. Experimental results underscore the superiority of the proposed HGNN-AR² in classification tasks compared to existing state-of-the-art methods. The primary contributions of this paper are listed as follows:

- A framework is proposed to address unestablished or inaccurate relations in constructing adjacency matrices for meta-paths by adaptively improving the graph structure generated by meta-path guidance.
- The framework harnesses the latent node representations derived from various meta-paths to enhance and rectify connections, thus efficiently optimizing the connectivity patterns within the graph structure.
- Experiments are performed on semi-supervised classification in four heterogeneous real-world datasets, and the results demonstrate competitive performance compared to existing state-of-the-art models.

The subsequent segments are organized as follows. In Section 2, we review the related works involved. Section 3 presents the definitions

of key terms and symbols used throughout the paper. In Section 4, we detail the proposed methodology, followed by an extensive experimental evaluation in Section 5. Eventually, the conclusions are presented in Section 6.

2. Related works

In this section, we will review and discuss some works related to graph neural networks and heterogeneous graph learning.

2.1. Graph neural networks

GNNs have received significant attention for their effectiveness in modeling complex graph structures. The primary principle of GNNs is to identify node relationships by investigating the overall graph structure, a capability that finds extensive application in various downstream tasks, such as node classification (Chen, Fu, Xiao, Wang, Plant, & Guo, 2024; Chen, Wu, Chen, Dong, & Wang, 2023), node clustering (Dor-naika & Hajjar, 2024; Fang, Du, Lin, Yang, Wang, & Shi, 2023), and link prediction (Bai, Nie, Zhang, Zhao, & Yuan, 2023; Zhao, Liu, Neves, Woodford, Jiang, & Shah, 2021).

Currently, GNNs can be categorized into two primary types, i.e., spectrum-based GNNs and spatial-based GNNs. The former utilizes the eigenvalue matrix of the graph Laplacian to conduct convolution operations in the Fourier domain of the graph. While the latter directly exploits node neighborhoods to perform convolution operations in the spatial nearest-neighbor group. Kipf and Welling (2017) proposed Graph Convolutional Network (GCN) bridging the spatial and spectral domains. GCN executes message aggregation in the spatial domain by approximating spectral graph convolution with a local first-order approach. Sun, Zhang, Chen, Xu, Zhang, and Yang (2023) introduced a flattened feature subspace and extended the feature space by incorporating structural principal components, improving the graph neural network architecture. Wang, Hu, Cao, Chang, and Tsang (2024) enhanced node representations by utilizing Laplacian disagreement to jointly assess initial features and output embeddings.

In the previously mentioned GNNs, a critical component is the message-passing mechanism, comprising two fundamental processes: aggregation of neighbor information and update of node representations. The complete information propagation mechanism can be expressed as follows:

$$\mathbf{h}_v^{(l+1)} = \sigma(\text{AGGREGATE}(\{\mathbf{h}_u^{(l)} : u \in \mathcal{N}(v)\})), \quad (1)$$

where $\mathbf{h}_v^{(l)}$ represents the representation of the v th node at the l th layer, $\mathcal{N}(v)$ denotes the set of the v th node neighbors, and $\sigma(\cdot)$ signifies the nonlinear activation function. The aggregation function AGGREGATE($\cdot$) collects information from neighboring nodes. This process enables nodes to propagate features through the graph structure, with node representations being updated by aggregating information from their neighbors. In this process, the aggregated representations of a node benefit from the features of neighboring nodes, provided these features contribute positively to the accuracy of the classification task.

However, this mechanism is designed for homogeneous graphs. Heterogeneous graphs, characterized by diverse node types, complex relationships, and limited node features, present challenges that hinder the direct application of these methods.

2.2. Meta-path based heterogeneous graph learning

In the realm of heterogeneous graph learning, meta-path based approaches have garnered considerable attention. Wang et al. (2019) incorporated meta-path and hierarchical attention mechanisms to formulate a heterogeneous graph neural network. Yu, Fu, Yu, Huang, Zhao, and Dong (2022) achieved the integration of multiple relationships among heterogeneous nodes, automatically capturing meta-path information of varying lengths by aggregating neighbor features. Yang,

Yan, Pan, Ye, and Fan (2023) aggregated heterogeneous nodes utilizing a lightweight mean aggregator and introduced a semantic fusion module based on transformers to fuse node representation. Lin et al. (2024) devised multiple meta-paths to construct neighborhood graphs and applied a hierarchical attention mechanism to aggregate node, path, and neighborhood graph information. Chen, Wu, et al. (2024) investigated the fusion of meta-paths of varied lengths, identifying finer-grained homomorphic connections between nodes through adaptive multi-order adjacency matrix learning.

While meta-paths offer a semantic interpretation of node relationships, they are insufficient to fully capture the complex semantic connections in heterogeneous graphs. Accurately depicting relationships between target nodes requires uncovering latent connections embedded within the topology and features of the graph.

2.3. Topology based graph augmentation

For node-level downstream tasks, topology based graph augmentation improves model generalization by denoising and enriching semantic relationships. Several methods (Wu, Chen, Du, Huang, & Wang, 2024; Zhao et al., 2021) achieve this by modifying the graph topology, either by adding or removing edges. Rong, Huang, Xu, and Huang (2020) introduced randomness and diversity into the input graph by randomly deleting edges from the original graph during each round of model training. Bo, Hu, Wang, Zhang, Shi, and Zhou (2022) proposed randomly replacing the direct neighbors of a node with remote neighbors and introduced neighborhood constraint regularization to ensure consistency in domain prediction, thereby enhancing data diversity. Shen, Sun, Pan, Zhou, and Yang (2023) employed a multi-head attention mechanism to automatically learn diverse graph-augmented views, improving contrastive learning for more effective node representation learning. Zhang, Xia, Cai, Yiu, Huang, and Jensen (2024) integrated the graph information bottleneck regularization paradigm to generate augmented graphs with enriched information for contrastive learning adaptively.

The negative impact of noisy or incomplete graph topologies can be reduced to some extent by using structural augmentation methods that are better suited to the requirements of downstream tasks.

3. Method motivation

3.1. Notation and definition

In this section, we introduce a few concepts and definitions of some terms that appeared in this paper.

Definition 1 (Heterogeneous Graph Zou, Fang, Wu, Zheng, & Wang, 2024). A heterogeneous graph, denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, associated with an entity type mapping function $\phi : \mathcal{V} \rightarrow \mathcal{A}$ and a link type mapping function $\psi : \mathcal{E} \rightarrow \mathcal{R}$. The sets $\mathcal{A}$ and $\mathcal{R}$ correspond to the predefined entity types and link types. $\mathcal{A}$ and $\mathcal{R}$, satisfying the condition $|\mathcal{A}| + |\mathcal{R}| > 2$.

Definition 2 (Meta-path Chen, Fuhai, Zhihao, Zhaoliang, Zhiling, Yan-chao, & Shipping, 2024).: A meta-path $\mathcal{P}$ is defined as the path in the form of $O_1 \xrightarrow{R_1} O_2 \xrightarrow{R_2} \dots \xrightarrow{R_{l-1}} O_l$ abbreviated as $O_1 O_2 \dots O_l$, which is the relation $R = R_1 \circ R_2 \circ \dots \circ R_{l-1}$ from entities O_1 to O_l , where $\circ$ is the composition operation on relations.

Definition 3 (Relation Homophily Tang, Chen, Wang, Xuan, & Zhao, 2023). Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where each node $v \in \mathcal{V}$ is associated with label y_v . Relation homophily h_r quantifies the extent to which a relation connects nodes of the same class and is formally defined as:

$$h_r = \frac{|\{(u, v) \in \mathcal{E} : y_u = y_v\}|}{|\mathcal{E}|},$$

Table 1

A summary of primary notations in this paper.

Notations	Explanations
$\mathbf{X} \in \mathbb{R}^{N \times F}$	The given heterogeneous feature matrix.
$\mathbf{A}_i \in \mathbb{R}^{N \times N}$	The i th heterogeneous meta-path matrix.
$\tilde{\mathbf{A}}_i \in \mathbb{R}^{N \times N}$	The i th reconstructed neighborhood matrix.
$\hat{\mathbf{A}}_i \in \mathbb{R}^{N \times N}$	The i th neighborhood matrix after sampling.
$\mathbf{Z}_i \in \mathbb{R}^{N \times d}$	The i th embedding matrix after transformation.
$\mathbf{H}_i \in \mathbb{R}^{N \times c}$	The i th representations of nodes.
$\mathbf{Y} \in \mathbb{R}^{N \times c}$	Real labels of the given target nodes.
$\hat{\mathbf{Y}} \in \mathbb{R}^{N \times c}$	Predictive labels of nodes after fusion.
$\mathbf{W} \in \mathbb{R}^{F \times d}$	Trainable network weight matrix.

Table 2

The performance (Macro-F1) of GCN in the classification task on the YELP dataset by manually defining different meta-paths and adjusting their connections.

Meta-path IDs	Original	Increase intra-class edge		Decrease inter-class edge	
		5%	10%	5%	10%
Meta-path 1	38.35	46.36	55.28	45.88	77.78
Meta-path 2	53.89	57.46	62.47	56.40	86.49
Meta-path 3	66.63	73.68	80.21	71.44	77.91
Average fusion	59.73	70.46	75.74	70.50	78.97
Weighted fusion	67.19	74.39	76.33	73.44	86.15

where $|\mathcal{E}|$ is the total number of relations (edges) and $|\{(u, v) \in \mathcal{E} : y_u = y_v\}|$ is the number of relations connecting nodes with the same label.

To facilitate comprehension of the mathematical symbols predominantly utilized in this paper, we provide explanations of these symbols in Table 1.

3.2. Experimental investigation

The performance of meta-path based heterogeneous graph models is highly contingent on the meta-path quality. We conduct simple experiments on the YELP dataset to elucidate these issues in Table 2. We utilize three manually defined meta-paths representing distinct semantic relationships. The performance of GCN is subpar when using these meta-paths. Given the substantial number of edges in YELP dataset, we increase intra-class edges by 5% or 10% and decrease inter-class edges by 5% or 10%. The manual adjustment results in significantly improved classification performance. This observation suggests that, despite the semantic justification for the employed meta-paths, the single meta-path or combination of multiple meta-paths fails to fully capture the underlying connectivity structure, which potentially impedes classification accuracy. We speculate the challenges in existing meta-path based methods, which may originate from the meta-paths themselves:

Unestablished Connection: The complexity of heterogeneous graphs poses a challenge in effectively exploring highly similar nodes within the actual topology using meta-paths. The representation of relationships among homomorphic nodes is often insufficient, failing to capture the full range of their relationships. Additionally, the relationships between nodes can be highly intricate, complicating the task of fully capturing their actual connectivity through manually defined meta-paths.

Inaccurate Connection: Despite the feasibility of manually defining meta-path definitions semantics, inaccuracies in connections still occur. For instance, a meta-path on Business–User–Business (BUB) on the YELP dataset represents connecting two businesses through users. The behavior of users may not necessarily reflect the business relationships, potentially resulting in the establishment of unnecessary connections between unrelated merchants.

The experimental results have demonstrated the significant impact of both intra-class and inter-class edges on classification performance. The impact can be elucidated through the message-passing mechanism of graph neural networks. Intra-class edges contribute to the

enhanced propagation of homophilic information, thereby improving the capability to learn features. This process further strengthens the intra-class correlation and similarity of nodes. Conversely, the presence of inter-class edges leads to information confusion or noise propagation between different classes, which potentially disrupts the ability of the network to learn correct classification patterns. Directly modifying connections based on labeling information is constrained by several factors. A primary limitation is the scarcity of sufficient node labels. Additionally, the semantic information conveyed through different meta-paths varies, resulting in partial discrepancies in the relational structures represented by specific connections within each meta-path.

To this end, we propose a model to address the aforementioned limitations. In order to complement and rectify connection relationships for different meta-paths, we first assess the similarity of the obtained latent connections between nodes. Subsequently, we estimate semantic connections for nodes under different meta-paths utilizing the measured similarity to capture diverse and sufficient connection patterns. Finally, the corrected adjacency structure undergoes graph convolution operations and fusion.

4. The proposed model

4.1. Relational probability prediction

Increasing intra-class edges while decreasing inter-class edges yields more efficient dissemination of node features, reinforcing the inclination towards class separation and yielding superior outcomes. One viable strategy involves constructing the graph topology based on node features to rectify neighbor relationships. However, in heterogeneous graphs, each meta-path encapsulates distinct semantic insights. To optimize complementary and corrective connections, an augmentation strategy is expected to be adopted to achieve an improved probability distribution of relationships.

Given the adjacency matrices $\{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_r\}$ obtained from r different meta-paths. Each meta-path encapsulates distinct semantic information within a heterogeneous graph and represents a specific topology. Excessive perturbation of the graph topology not only undermines robustness but also impairs generalizability. A combination of graph topology and feature information is employed to extract preliminary node representations and map them to the latent variable distribution space. For each adjacency matrix $\mathbf{A}_i$, we employ a GCN as the encoder layer, with the formula as follows:

$$\mathbf{Z}_i = \mathcal{F}_{GCN}(\mathbf{X}, \mathbf{A}_i; \{\mathbf{W}_l^{(l)}\}_{l=1}^L), \quad (2)$$

where $\mathbf{Z}_i$ denotes the node embedding within a specific topology $\mathbf{A}_i$. Through the encoder layer, the initial node features are propagated under the guidance of the adjacency matrix $\mathbf{A}_i$. In this process, node features are pre-integrated based on the semantic relationships represented by the meta-paths, facilitating the separation of nodes across different classes.

To evaluate the homomorphic trend in latent node representations, node connectivity is predicted by projecting the latent vector $\mathbf{Z}_i$ into a relational probability space:

$$\tilde{\mathbf{A}}_i = \text{Norm}(\mathbf{Z}_i \mathbf{Z}_i^T), \quad (3)$$

where $\tilde{\mathbf{A}}_i$ denotes the predicted edge probability matrix generated by the inner product decoder for the i th element, $\text{Norm}(\cdot)$ denotes the normalization process that maps the dense probability matrix to the adjacency space of the graph so that it conforms to the adjacency structure.

The encoding phase accentuates the discrepancies between nodes of different classes and strengthens the neighborhood information derived from meta-paths. Meanwhile, the decoding phase transforms the node embeddings into predicted connection values, thereby extracting insights into the homomorphic relationships among the nodes.

4.2. Sparse relation sampling

We enhance the adjacency relationships reflected in different meta-paths by extracting latent connections and similarity metrics. However, we obtain a dense representation where each node pair is associated with a probability. To preserve storage and computational resources, we introduce sparsification and sampling operations.

We anticipate that the reconstructed neighborhood will enhance potential connections among similar nodes. However, this fails to imply discarding the original connections to accommodate other types of connections. Therefore, it is essential to ensure that the connection distribution generated by the edge predictor is consistent with the original neighborhood structure. For each generated $\tilde{\mathbf{A}}_i$, we interpolate it with the original $\mathbf{A}_i$ to produce the fused neighborhood relation $\mathbf{P}_i$. In particular, for r meta-paths, we obtain r fused adjacency matrices $\mathbf{P}_i$, denoted as $\{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_r\}$, where the values in $\mathbf{P}_i$ indicate the probabilities of edge existence under the i th meta-path.

$$\mathbf{P}_i = \alpha \tilde{\mathbf{A}}_i + (1 - \alpha) \mathbf{A}_i, \quad (4)$$

where α is a hyperparameter acting as the equalization factor. As each edge prediction probability matrix $\mathbf{P}_i$ is dense, even though the likelihood of intra-class edges is low, they still influence the message propagation. We utilize Bernoulli sampling to remove edges with low probabilities. Since Bernoulli sampling is inherently a discrete operation, we employ a soft, continuous relaxation to approximate it.

Consider a binary variable as the presence of an edge in a graph reconstruction. This variable can be characterized by a Bernoulli distribution, with $\mathbf{p}$ denoting the probability of an edge. To facilitate differentiable sampling, we employ the Gumbel-Softmax reparameterization (Jang, Gu, & Poole, 2017) as follows.

For each binary variable $\mathbf{x}_{ij}$, we generate Gumbel noise $\mathbf{z}_{ij}$:

$$\mathbf{z}_{ij} = -\log(-\log(\mathbf{U}_{ij})), \quad (5)$$

where $\mathbf{U}_{ij}$ represents a sample drawn from the uniform distribution $U[0, 1]$. Subsequently, we employ the Softmax function to map the Gumbel noise to the parameter space of the Bernoulli distribution:

$$\text{softmax}(\mathbf{z}_{ij}) = \frac{e^{(\log(\mathbf{p}_{ij}) + \mathbf{z}_{ij})/\tau}}{e^{(\log(\mathbf{p}_{ij}) + \mathbf{z}_{ij})/\tau} + e^{(\log(1 - \mathbf{p}_{ij}) + \mathbf{z}_{ij})/\tau}}, \quad (6)$$

where τ denotes the temperature coefficient. Throughout the training process, we interpret each output of $\text{softmax}(\mathbf{z}_{ij})$ as the probability associated with the Bernoulli distribution from which we perform sampling.

We employ relaxed sampling and utilize a gradient estimator called straight-through (ST) (Bengio, Léonard, & Courville, 2013). This estimator operates by rounding the relaxed samples during the forward pass and directly propagating the gradient through the relaxed samples during the backward pass. It is formulated as:

$$\hat{\mathbf{A}}_{ij} = \left\lfloor \frac{1}{(1 + e^{-\log(\mathbf{P}_{ij} + G)/\tau})} + \frac{1}{2} \right\rfloor, \quad (7)$$

where $G \sim \text{Gumbel}(0, 1)$ is a Gumbel random variate. We condense the dense edge probability matrix into the graph topology that encapsulates homomorphic connections and sidesteps the eradication of other relations.

4.3. Inference and optimization

After applying the components introduced in the preceding sections, we obtain r different reconstructed adjacency matrices. Each reconstructed adjacency matrix $\hat{\mathbf{A}}_i$, along with the node features $\mathbf{X}$, is passed to the graph node classifier to obtain node representations $\mathbf{H}_i$ for each channel. We design a learnable weight $\{\omega_r\}_{i=1}^r$ with $\sum_{i=1}^r \omega_i = 1$ to combine the different representations, ultimately yielding the representation $\mathbf{H}$ of the heterogeneous graph nodes, expressed as:

$$\mathbf{H} = \sum_{i=1}^r \omega_i \mathbf{H}_i, \quad (8)$$

Algorithm 1 Heterogeneous Graph Neural Network with Adaptive Relation Reconstruction (HGNN-AR²)

Input:

Target node feature matrix $\mathbf{X}$,
Ground truth $\mathbf{Y}$,
Hyperparameters α, β , and τ .

Output: Predictive label $\hat{\mathbf{Y}}$.

```

1: Initialize weights and biases;
2: Generate  $\{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_r\}$  based on different meta-paths;
3: while not convergent do
4:   for  $i = 1 \rightarrow r$  do
5:     Compute the reconstruction matrix  $\tilde{\mathbf{A}}_i$  by utilizing the learned
       node embeddings via Equation (3);
6:     Determine matrix  $\hat{\mathbf{A}}_i$  by conducting interpolation and sampling
       operations via Equation (7);
7:   end for
8:   Evaluate  $\omega_r$  and  $\mathbf{H}$  and by Equation (8);
9:   Calculate loss value  $\mathcal{L}$  via Equations (11);
10:  Optimize the trainable parameters by backward propagation;
11: end while
12: Generate label  $\hat{\mathbf{Y}}$  based on the predictive output  $\mathbf{H}$ ;
13: return Predictive label  $\hat{\mathbf{Y}}$ .
```

Through the above process, the model effectively refines the adjacency relationships constructed by each meta-path, ensuring the feature propagation process within the graph neural network is constrained to nodes of the same class. This constraint enables the model to produce more comprehensive and accurate node representations. Subsequently, the model utilizes the combination of these feature representations for prediction.

Here, we use logarithmic likelihood reconstruction error to assess the loss $\mathcal{L}_{rec}$ of the edge prediction module, defined as:

$$\mathcal{L}_{rec} = \sum_{i=1}^r \left(-\frac{1}{\kappa_i} \sum_{i,j} \mathbf{A}_{ij}^{(r)} \log(\hat{\mathbf{A}}_{ij}^{(r)}) \right), \quad (9)$$

where κ_i is the number of non-zero entries in $\mathbf{A}_i$. The loss $\mathcal{L}_{ce}$ of the node classification module is evaluated through the cross-entropy error of all labeled samples, defined as:

$$\mathcal{L}_{ce} = - \sum_{i \in \Omega} \sum_{j=1}^c \mathbf{Y}_{ij} \log(\hat{\mathbf{Y}}_{ij}), \quad (10)$$

where $\hat{\mathbf{Y}} = \text{softmax}(\mathbf{H})$, and c is the number of classes.

Subsequently, we conduct backpropagation using a joint cross-entropy loss $\mathcal{L}_{ce}$ and an edge reconstruction loss $\mathcal{L}_{rec}$:

$$\mathcal{L} = \mathcal{L}_{ce} + \beta \mathcal{L}_{rec}, \quad (11)$$

where β is the equilibrium parameter that regulates the loss weight of the reconstruction.

We introduce an end-to-end trainable methodology wherein the edge prediction and classification modules undergo co-training. This design enables exploring extensive connectivity information by leveraging the labeled data from the classifier to guide the graph reconstruction. Continuous training leads to more comprehensive target node representations within the heterogeneous graph. The specific process of the proposed method is summarized in Algorithm 1.

4.4. Time complexity analysis

The time complexity of the proposed HGNN-AR² is primarily determined by two phases: relational reconstruction and sparsification sampling.

First, a graph convolution layer is employed as an encoder to generate node embeddings, with the time complexity of the graph

Table 3

A elaborate description of the test heterogeneous datasets..

Datasets	Samples	Features	Classes	Meta-Paths	Homophily
ACM	3,025	1,870	3	PAP	0.85
				PSP	0.65
DBLP	4,057	334	4	APA	0.92
				APVPA	0.82
				APTPA	0.60
IMDB	4,780	1,232	3	MAM	0.57
				MDM	0.75
				MYM	0.51
YELP	2,614	82	3	BUB	0.68
				BLB	0.77
				BSB	0.68

convolution operation being $O(mN^2d)$, where m denotes the number of meta-paths, N represents the number of nodes and d denotes the embedding dimension. Following the generation of node embeddings, the graph structure is reconstructed by computing the outer product of the embeddings, which incurs a time complexity of $O(mN^2d)$. This reconstructed adjacency matrix is then normalized, resulting in an additional complexity of $O(mN^2)$. Subsequently, another graph convolutional network is used for classification, with the time complexity again being $O(mN^2d)$. Finally, predicted labels are generated via a softmax operation, with a time complexity of $O(Nc)$, where c is the number of output classes. The total time complexity can thus be expressed as $O(mN^2d) + O(mN^2d) + O(mN^2) + O(mN^2) + O(mN^2d) + O(Nc)$, which simplifies to $O(mN^2d)$.

5. Experiments

In this section, we conduct comprehensive experiments to demonstrate the effectiveness of the proposed method. We adopt four widely used heterogeneous graph datasets to evaluate the performance of HGNN-AR² relative to classical and state-of-the-art baselines.

5.1. Experimental setups

Datasets. We adopt four publicly available datasets to evaluate the performance of compared methods, i.e., ACM, DBLP, IMDB, and YELP. Detailed descriptions are provided as follows.

- **ACM** is an academic citation network containing three types of nodes: papers (P), authors (A), and study subjects (S), and we employ the meta-path set $\{PAP, PSP\}$ for experiments.
- **DBLP** is extracted from the citation site, which contains four types of nodes: papers (P), authors (A), venues (V), and terms (T). To conduct experiments, we generate the meta-path set $\{APA, APVPA, APTPA\}$.
- **IMDB** is derived from an online movie database. We extracted a subset containing a subset of four types of nodes: movies (M), actors (A), directors (D), and years (Y). We conduct experiments by the meta-path set $\{MAM, MDM, MYM\}$.
- **YELP** is a heterogeneous dataset derived from a consumer review website. We adopt the subset of YELP that comprises four types of nodes: businesses (B), users (U), services (S), and levels (L). The meta-path set $\{BUB, BSB, BLB\}$ are employed to conduct experiments.

A brief description of these datasets is summarized in Table 3.

Baselines. In this section, we perform an exhaustive comparative analysis of the proposed method by evaluating it comprehensively with nine baseline models, which are given below.

- **GCN** (Kipf & Welling, 2017) is a homogeneous graph neural network. We target adjacency matrices generated by different meta-paths fed into GCN module and aggregate the resulting representation through summation.
- **HAN** (Wang et al., 2019) is designed for heterogeneous graphs that learn representation based on different meta-paths by introducing an attention mechanism that combines these meta-path-specific representations.
- **MRGCN** (Huang, Li, Ye, & Ng, 2020) introduces a convolution operator based on graph spectral analysis to capture correlations across relations effectively, forming a multi-relational graph neural network framework.
- **SSDCM** (Mitra, Vijayan, Singh, Goswami, Parthasarathy, & Ravindran, 2021) is a multiplexed network structure-aware representation learning method designed to maximize the mutual information between local and contextualized global graph summaries.
- **MHGCN** (Yu et al., 2022) is a multiplexed heterogeneous graph convolutional network that learns representation through aggregating different meta-paths and capturing beneficial relationship-aware topology.
- **HGATE** (Wang et al., 2023) extends the graph auto-encoder to accommodate heterogeneous graphs, employing hierarchical attention to learn correlations between nodes and their meta-path based neighbors.
- **MGDCR** (Mo et al., 2023) introduces a self-supervised multiplex graph representation learning framework that mitigates noise within each graph structure through a dual decorrelation loss.
- **HMGE** (Abdous, Mrabah, & Bouguessa, 2024) employs hierarchical aggregation to progressively combine multiple relational dimensions of the graph, capturing complex interactions and revealing latent structures through non-linear combinations.
- **AMOGCN** (Chen, Wu, et al., 2024) constructs a fused multi-order adjacency matrix and automatically explores meta-paths involving multi-hop neighbors under the guidance of node semantic information.

5.2. Semi-supervised classification

In this section, we evaluate the effectiveness of our model in comparison to state-of-the-art methods in node classification. We select Macro-F1 and Micro-F1 as the evaluation metrics and employ the Adam optimizer. All methods undergo 5 repetitions, and we record average performance. We randomly partition all datasets into varying proportions of training sets (20%/40%/60%). The experimental results are presented in Table 4.

Specifically, with labeled data comprising 20%, our model demonstrates superior performance across diverse datasets. Taking the Macro-F1 evaluation metric as an example, our method outperforms the runner-up by 0.70% on the ACM dataset, achieves a 0.50% enhancement on the DBLP dataset, observes a 1.59% improvement on the IMDB dataset, and records a 2.69% enhancement on the YELP dataset. Notably, our model maintains its leading position even with a higher proportion of labeled data. In summary, our model achieves superior performance on these four datasets, which significantly demonstrates its effectiveness.

5.3. Robustness analysis

To evaluate the robustness of the model, we randomly introduce varying proportions of noisy edges into the meta-path-based topology and observe the corresponding performance variations. As illustrated in Fig. 3(a), the performance of all algorithms exhibits a declining trend with the increase in noise ratio. Nevertheless, the proposed HGNN-AR² demonstrates significantly greater robustness and less performance degradation compared to the baseline algorithms. Even under a high

Table 4
Node classification performance for multiple training sample percentages..

Datasets	Metric	Training	GCN	HAN	MRGCN	SSDCM	MHGCN	HGATE	MGDCR	HMGE	AMOGCN	HGNN-AR ²
ACM	Macro-F1	20%	78.62	88.03	87.64	87.72	88.96	88.73	89.03	90.80	<u>91.72</u>	92.42
		40%	75.18	91.03	88.49	88.12	91.10	89.16	90.32	92.45	<u>92.40</u>	92.85
		60%	76.44	89.21	91.57	88.69	93.73	91.08	93.95	91.20	<u>94.06</u>	94.36
	Micro-F1	20%	78.88	88.06	87.52	87.65	89.19	88.70	88.87	90.66	<u>91.39</u>	92.36
		40%	75.86	91.16	88.37	88.32	91.81	89.04	90.37	92.35	<u>92.38</u>	92.93
		60%	75.59	89.16	91.46	88.83	92.45	91.03	93.89	90.96	<u>94.04</u>	94.40
DBLP	Macro-F1	20%	90.15	89.30	89.53	89.44	90.90	90.25	92.04	91.56	92.02	92.54
		40%	89.56	90.00	91.22	90.27	89.56	91.33	91.82	91.89	<u>92.07</u>	93.06
		60%	90.25	90.74	91.13	90.96	92.23	92.50	91.73	91.48	<u>93.56</u>	93.73
	Micro-F1	20%	91.63	90.43	90.51	89.94	92.17	91.15	92.43	92.35	<u>92.46</u>	93.02
		40%	90.67	90.42	91.72	90.61	90.48	91.73	91.89	92.31	<u>92.36</u>	93.44
		60%	90.96	91.17	91.65	91.10	93.62	92.20	92.73	92.17	<u>93.74</u>	94.15
IMDB	Macro-F1	20%	24.43	49.83	45.62	49.45	50.54	41.99	45.03	32.93	<u>50.92</u>	52.51
		40%	24.47	52.04	45.70	52.21	52.34	48.28	47.42	36.46	52.27	53.23
		60%	28.79	54.22	49.29	53.90	53.17	51.28	50.25	40.30	52.65	<u>54.12</u>
	Micro-F1	20%	55.46	54.92	47.78	59.16	64.20	52.10	59.05	57.35	<u>63.60</u>	61.36
		40%	54.46	54.10	48.31	59.32	61.32	56.09	60.11	57.29	<u>61.72</u>	63.04
		60%	50.54	56.69	51.84	60.13	61.54	61.55	59.62	60.36	<u>62.85</u>	64.13
YELP	Macro-F1	20%	52.05	59.98	54.48	53.74	74.93	<u>91.41</u>	78.18	58.57	71.64	93.77
		40%	53.09	60.16	54.71	71.18	76.17	<u>92.54</u>	77.65	55.53	70.43	94.39
		60%	58.08	61.32	53.57	75.94	74.05	<u>92.53</u>	80.73	55.66	71.58	94.51
	Micro-F1	20%	67.49	73.53	73.70	72.71	78.61	<u>90.42</u>	81.57	74.84	76.63	93.11
		40%	71.35	74.36	73.59	77.02	80.36	<u>91.57</u>	81.23	75.48	78.54	93.94
		60%	73.63	73.43	72.62	79.27	80.12	<u>91.57</u>	82.72	75.48	78.16	93.99

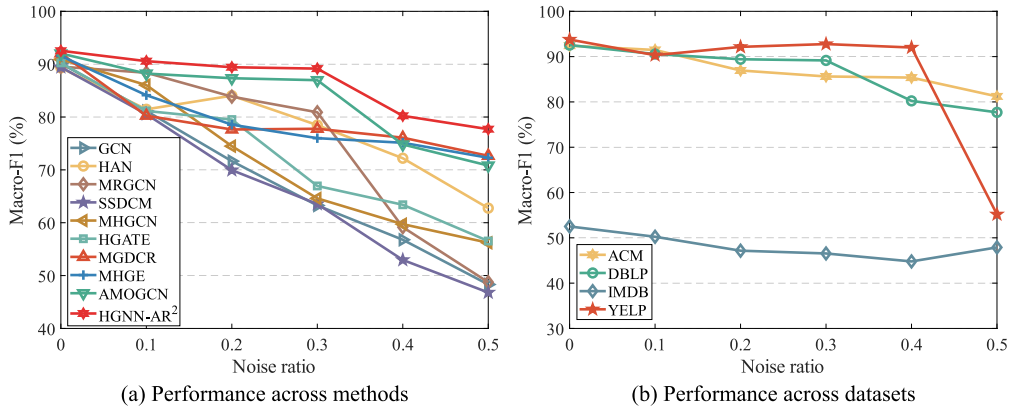


Fig. 3. Robustness analysis with varying added noise ratios: (a) Performance of different methods on DBLP; (b) Robustness of the proposed HGNN-AR² across four datasets.

noise ratio of 0.5, the proposed HGNN-AR² consistently achieves superior performance relative to the baseline methods. Fig. 3(b) shows that HGNN-AR² exhibits better robustness across most datasets, with performance degradation occurring more gradually. It is noteworthy that the YELP dataset experiences a sharp decline in performance at a higher noise ratio of 0.5. We hypothesize that this may be due to the higher number of inter-class edges in the YELP dataset, which could also explain the relatively poor performance of some baseline methods. Overall, these results demonstrate that HGNN-AR² achieves better robustness, further validating the effectiveness of the proposed method.

5.4. Data visualization

Heatmap Visualization. We present the reordered one of the relation prediction probability matrix $\hat{A}$ in the ACM and DBLP datasets, respectively, with the corresponding results depicted in Fig. 4. We can observe a distinct block structure, which signifies that the reconstructed nodes within the same class possess significantly higher weights compared to those between different classes, thereby exhibiting enhanced heteromorphic correlation. Through the refinement of

node connections, the connection deficiencies and heteromorphism limitations associated with the mitigated meta-path can be effectively alleviated.

T-SNE Visualization. To offer a more intuitive assessment of our framework, we opt to visualize the node representations using the DBLP dataset. The results are depicted in Fig. 5. Most benchmark methods demonstrate strong classification performance with 20% of labeled data. For instance, algorithms like MGDCR and AMOGCN exhibit relatively clear class boundaries; however, nodes within the same class tend to form multiple distinct clusters. This phenomenon indicates that these algorithms struggle to preserve feature correlation among nodes of the same class, potentially limiting their ability to capture intra-class consistency.

In contrast, HGNN-AR² demonstrates more distinguishable boundaries, with heightened consistency among nodes, leading to more separable node clusters. HGNN-AR² promotes the propagation of intra-class node features while weakening the propagation of inter-class node features, consequently enhancing the consistency of embeddings for nodes within the same class. This observation highlights the superior node representation capability of the proposed method, affirming its effectiveness.

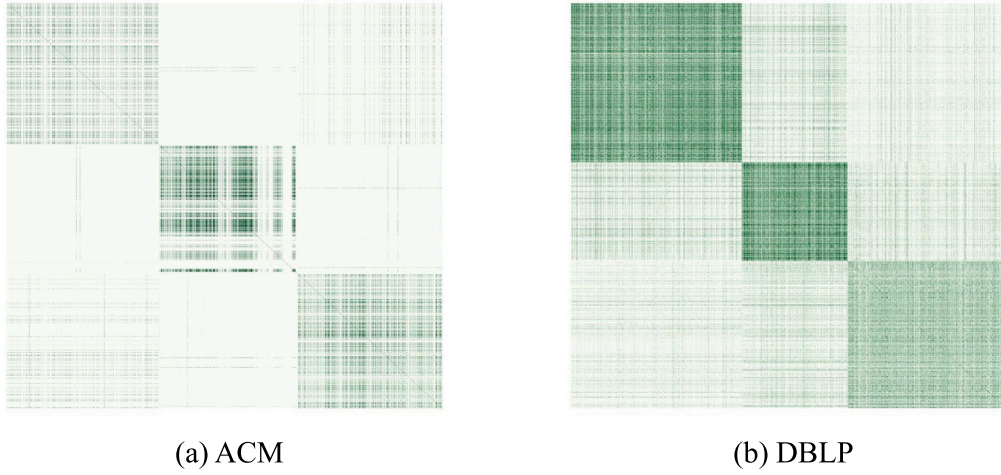


Fig. 4. (a) and (b) indicate the heatmap of the reconstructed relation probability matrix for the ACM and DBLP datasets reordered by node labels.

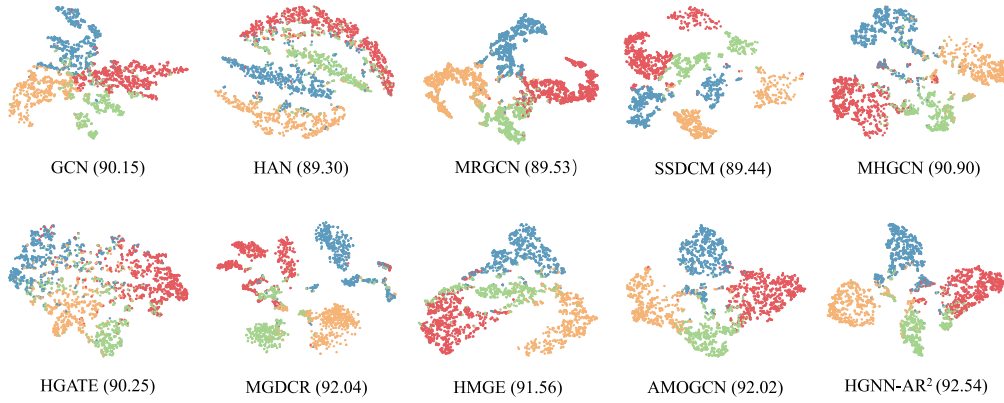


Fig. 5. T-SNE Visualization and Macro-F1 values of compared methods on the DBLP dataset with 20% training samples.

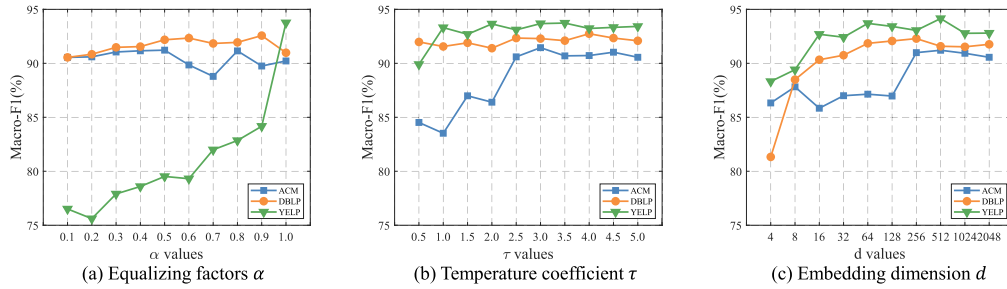


Fig. 6. The classification accuracy of the proposed method w.r.t.: (a) equalizing factor α ; (b) temperature coefficient τ ; (c) embedding dimensions d .

5.5. Parameter sensitivity

In this section, we conduct sensitivity analyses on the proposed method parameters to validate their effectiveness. Given the significant variability in accuracy across different datasets, we focus on three datasets, ACM, DBLP, and YELP, for parameter sensitivity analyses.

Effect of the equalizing factor α : The precise setting of α determines the equilibrium between preserving the original topology and revealing intricate connections. As shown in Fig. 6(a), excessively large or small values of α impact performance. This situation might occur because of the plethora of inter-class connections derived from meta-paths. This abundance introduces noise into the reconstructed adjacency matrix, leading to a significant decline in performance.

Effect of the temperature coefficient τ : The temperature coefficient τ serves as a regulator in the Bernoulli sampling process. Elevated temperatures heighten the probability of selecting minor probability values, whereas lower temperatures make the sampling more deterministic. The findings in Fig. 6(b) demonstrate a consistent variety. Increasing the value of τ enhances the diversity within the adjacency matrix, thereby uncovering additional intra-class connections.

Effect of the embedding size d : From the results of Fig. 6(c), we discern that an embedding size of 256 achieves better classification performance. A low-dimensional embedding matrix might fail to capture crucial information, thereby hindering the accurate representation of the graph structure. Heightened dimensionality may exacerbate overfitting, thereby diminishing the effectiveness of the model.

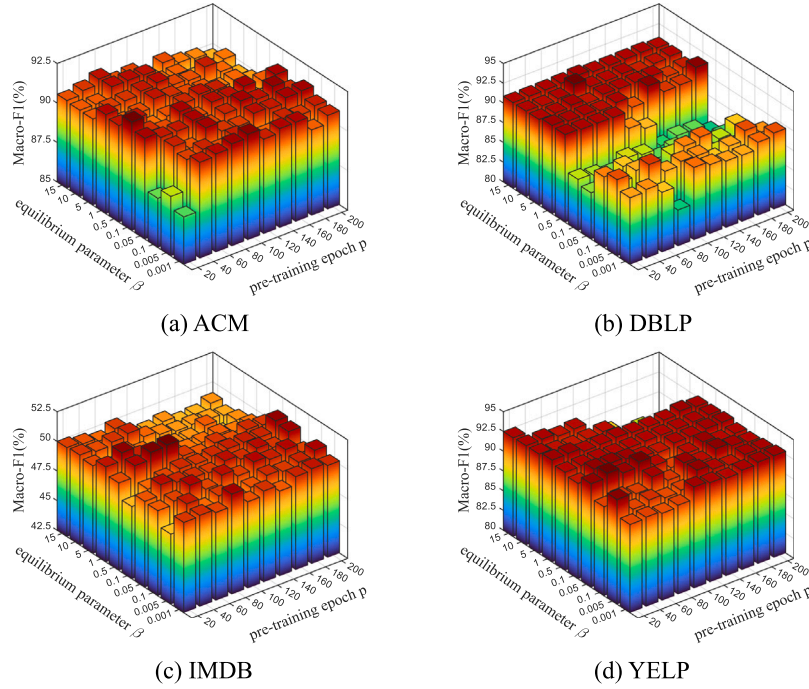


Fig. 7. Parameter sensitivity analysis concerning the equilibrium parameter β and the pre-training epoch p of the proposed method.

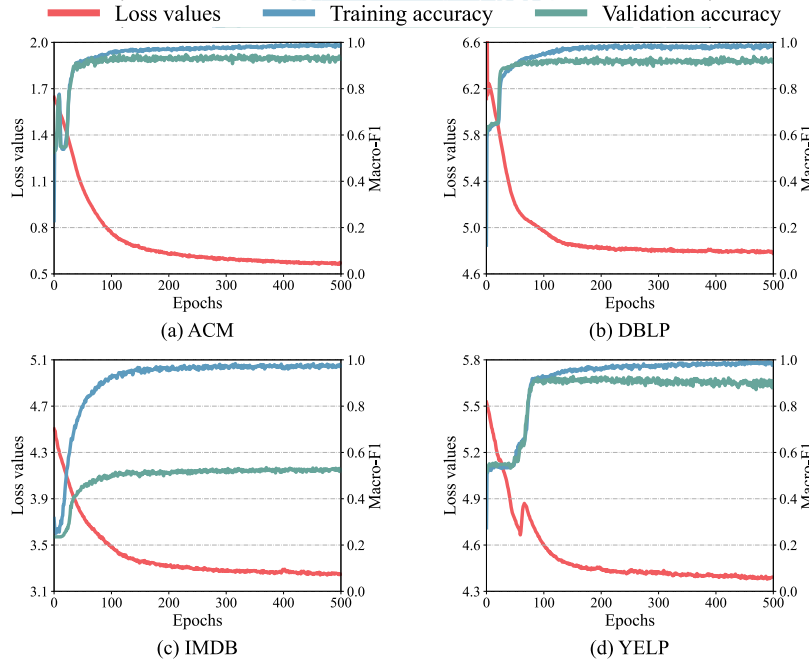


Fig. 8. Convergence curves of the proposed method on four tested datasets, where the green curve indicates the Macro-F1 on the validation set, and the orange curve indicates the loss variation.

Effect of the equilibrium parameter β and the pre-training epoch p : In Fig. 7, we examine the impact of the equilibrium parameter β and the number of pre-training epoch p of graph auto-encoder on model performance. The analysis reveals when the number of pre-training epochs is held constant, excessive and insufficient values of β result in decreased performance. Excessive values of these parameters may induce overfitting during the encoding phase, causing the embeddings of different node classes to exhibit homogeneity tendencies.

Insufficient value fails to effectively distinguish between embeddings of different classes.

5.6. Convergence analysis

In this section, we document the loss and accuracy curves on the validation set throughout the learning process. As shown in Fig. 8, the loss curves exhibit a sharp downward trend across datasets, with both the loss value and Macro-F1 score gradually stabilizing as

Table 5

Ablation study of the proposed model on all test datasets.

Methods/Datasets	ACM		DBLP		IMDB		YELP	
	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1
Model-1 w/o GAE	82.76	82.51	88.32	89.32	50.44	58.09	67.19	77.19
Model-2 w/o $\mathcal{L}_{rec}$	88.80	88.75	89.33	90.27	49.83	56.98	71.31	77.90
Model-3 w/o Sparsification	77.23	76.60	90.89	91.47	50.60	57.28	69.92	79.26
Model-4 w/o Bernoulli Sampling	90.70	90.69	88.29	89.15	48.01	55.61	90.49	89.72
HGNN-AR ²	92.42	92.36	92.54	93.02	52.51	61.36	93.77	93.11

training epochs increase, showing minimal fluctuations, and ultimately converging after 500 epochs.

Analyzing the convergence curve of the proposed method on the YELP dataset, we notice significant fluctuations in the loss, accompanied by a brief surge during convergence, but eventually level off.

5.7. Ablation study

To assess the effectiveness of each model component, we conduct an ablation study to examine the effect of different components. This involves evaluating the model performance across four datasets, with the findings detailed in Table 5.

Model-1 w/o GAE: We remove the edge prediction module while keeping the topology of the meta-path-based graph unchanged. The original adjacency matrix is directly fed into the classifier without any modifications.

Model-2 w/o $\mathcal{L}_{rec}$: We set $\beta = 0$ to isolate the impact of the reconstruction loss on the overarching model dynamics. When the reconstruction loss $\mathcal{L}_{rec}$ is decoupled from the cross-entropy loss $\mathcal{L}_{ce}$ during training, there exists a risk of the generated outputs diverging from the original neighborhood graph, potentially introducing instability to the training process.

Model-3 w/o Sparsification: We intentionally bypass the ensuing sparsification and directly feed the dense neighborhood into the classifier. This configuration facilitates the assignment of distinct weights for each edge. These parameters are collectively utilized in convolution.

Model-4 w/o Bernoulli Sampling: In our endeavor to assess the implications of the sampling process, we consciously abstain from employing Bernoulli sampling. We adopt a deterministic approach, rounding the edge probabilities to binary values (0 or 1). This deterministic rounding may reduce diversity during model training, as some edges might be overlooked due to the influence of the original structure, ultimately leading to a decline in model performance.

6. Conclusion

In this paper, we explored the problem of unestablished and heterogeneous connections in meta-path based heterogeneous graph learning methods based on meta-paths and their impact on node classification accuracy. We proposed a heterogeneous graph neural network model to alleviate these issues. Grounded on different connections derived from multiple meta-paths, the model reshaped the appropriate adjacency relationships by examining the homogeneous correlations of node latent features under specific connections to generate more comprehensive representations. In future work, we will focus on exploring more flexible and scalable model architectures, refining meta-paths to more effectively capture the complex and dynamic relationships between nodes, and improving the ability to represent intricate dependencies within graph structures.

CRedit authorship contribution statement

Weihong Lin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Zhaoliang Chen:** Writing – review & editing, Supervision, Project administration,

Methodology, Formal analysis, Data curation. **Yuhong Chen:** Writing – review & editing, Supervision, Resources, Formal analysis, Data curation. **Shiping Wang:** Writing – review & editing, Supervision, Resources, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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